FUNCTIONS

CHAPTER 6

Rational and Exponential Functions

6

SYLLABUS 1.11 2.8 2.9 2.13

Pour a cup of coffee at $90\,^{\circ}$C and set it down in a room at $20\,^{\circ}$C. It cools quickly at first, then more slowly, then so slowly that you can barely measure the change. Leave it long enough and it reaches room temperature. That is what we usually say. Now look at the numbers instead: the difference between the coffee and the room keeps halving. It becomes smaller and smaller, but it never becomes zero.

Isaac Newton measured this and found the rule. In each fixed interval of time, the temperature difference falls by the same *fraction*, not by the same amount. That property is what defines an **exponential**. Plot the temperature against time and the curve falls steeply, then bends and flattens, settling towards the horizontal line at $20\,^{\circ}$C without ever reaching it.

That line is an **asymptote**, and asymptotes appear in every section of this chapter. A curve can approach a value it never reaches. It can also increase without bound as its input approaches a value where the function is undefined. Reading that behaviour from the equation alone is the skill this chapter builds.

Two families of function behave this way. **Rational** functions are ratios of polynomials. The **transcendental** pair are the exponential and the logarithm, each the mirror image of the other in the line $y = x$. Between them they model cooling coffee, growing populations, decaying atoms and compound interest. By the end of the chapter you will be able to sketch any of them from its formula.

More precisely, you will find the asymptotes of a rational function and sketch it, handle an oblique asymptote by division, treat the exponential and the logarithm as inverses, and split a rational expression into partial fractions. These functions earn their place because they model what lines and parabolas cannot. One grows in proportion to its own size; another approaches a limit it never reaches.

6.1 Reciprocal and Rational Functions

The simplest function with an asymptote is the **reciprocal** $y = \dfrac{1}{x}$.

It is undefined at $x = 0$. As x approaches 0, the output grows without bound, towards $+\infty$ on one side and $-\infty$ on the other. For large $|x|$ the opposite happens: the output becomes very small, and the curve approaches zero.

So the reciprocal has two asymptotes: the vertical line $x = 0$ and the horizontal line $y = 0$. It is also **self-inverse**, meaning that applying it twice returns the original input, so its graph is symmetric in the line $y = x$. See Ch. 5.

It is defined for every input except $x = 0$, so its **domain** is all real numbers with $x \neq 0$. It produces every output except 0, so its **range** is all real numbers with $y \neq 0$.

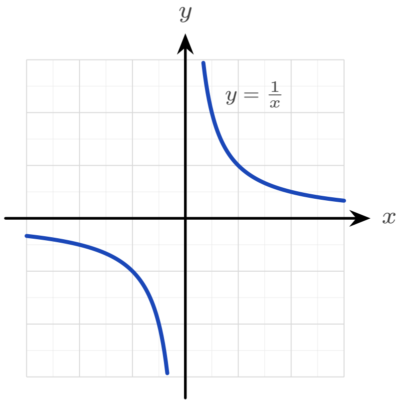

Figure 6.1 The reciprocal $y = \frac{1}{x}$. Its asymptotes are the axes themselves: the curve approaches $x = 0$ and $y = 0$ without meeting either. The domain is every real number except 0; the range is every real number except 0.

> DEF An **asymptote** is a line the graph of a function approaches arbitrarily closely. A **vertical asymptote** occurs at a value of x where the function is undefined and grows without bound. A **horizontal asymptote** is a value the function approaches as $x \to \pm\infty$.

Every function of the form $\dfrac{ax+b}{cx+d}$, with $c \neq 0$ and $ad - bc \neq 0$, is a reciprocal curve that has been shifted and stretched. It has one vertical asymptote, where the denominator is zero, and one horizontal asymptote, at the ratio of the leading coefficients.

Both conditions matter. If $c = 0$ the function is linear and has no asymptote at all; if $ad - bc = 0$ the numerator is a multiple of the denominator and the function is constant.

KEY For $f(x) = \dfrac{ax+b}{cx+d}$, with $c \neq 0$ and $ad - bc \neq 0$:

$$\text{vertical asymptote at } x = -\frac{d}{c}, \qquad \text{horizontal asymptote at } y = \frac{a}{c}.$$

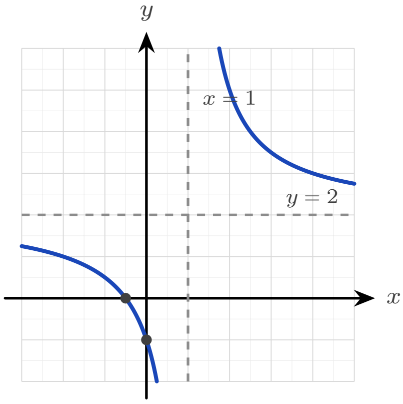

Figure 6.2 The curve $y = \frac{2x+1}{x-1}$, with its two asymptotes dashed and its two intercepts marked. These four features are what a sketch must show.

EXAMPLE 6.1

Sketch $f(x) = \dfrac{2x+1}{x-1}$, showing all asymptotes and intercepts.

Find the vertical asymptote first. The denominator is zero at $x = 1$, so the line $x = 1$ is the vertical asymptote.

Next the horizontal asymptote. The ratio of the leading coefficients gives $y = \dfrac{2}{1} = 2$.

Rewriting the function confirms the shape. Divide the numerator by the denominator:

$$\frac{2x+1}{x-1} = 2 + \frac{3}{x-1},$$

a reciprocal curve centred on the crossing of the two asymptotes, $(1, 2)$.

Now find the intercepts. Substitute $x = 0$ for the y-intercept: $f(0) = \dfrac{1}{-1} = -1$. Set the numerator to zero for the x-intercept: $2x + 1 = 0$, so $x = -\frac{1}{2}$. Both are marked on the figure above.

Two asymptotes and two intercepts fix the whole sketch.

TIP A sketch question is marked on exactly four features: both asymptotes and both intercepts. Label each one on the diagram; an unlabelled curve of the right shape scores very little.

YOUR TURN 6.1 Reciprocal and Rational Functions

1. State the vertical asymptote of each function.

(a) $f(x) = \dfrac{1}{x-3}$ [1] (b) $f(x) = \dfrac{1}{x-5}$ [1]

(c) $f(x) = \dfrac{3}{x+2}$ [1] (d) $f(x) = \dfrac{2x}{x+1}$ [1]

2. State the horizontal asymptote of each function.

(a) $f(x) = \dfrac{2x}{x+1}$ [1] (b) $f(x) = \dfrac{x+1}{x-2}$ [1]

(c) $f(x) = \dfrac{3x-6}{x+1}$ [2] (d) $f(x) = \dfrac{3}{x+2}$ [2]

3. State both asymptotes and both intercepts of each function.

(a) $f(x) = \dfrac{x-1}{x+2}$ [4]

(b) $f(x) = \dfrac{2x+3}{x-1}$ [4]

4. State the domain and range of each function.

(a) $f(x) = \dfrac{1}{x-5}$ [2] (b) $f(x) = \dfrac{1}{x} + 4$ [3]

5. Consider $f(x) = \dfrac{3x-6}{x+1}$.

(a) Write $f(x)$ in the form $A + \dfrac{B}{x+1}$. [3]

(b) Hence state the point on which the two asymptotes cross. [2]

(c) Sketch the curve, showing both asymptotes and both intercepts. [3]

6.2 Rational Functions and Oblique Asymptotes HL

When the numerator has a higher degree than the denominator, there is no horizontal asymptote. The curve approaches a slanted line instead.

The useful case is when the numerator's degree is exactly one more than the denominator's. Polynomial division then splits the function into a linear part plus a remainder, and that remainder approaches zero for large $|x|$. The linear part is the **oblique** (slant) asymptote.

EXAMPLE 6.2

Find the asymptotes of $f(x) = x + \dfrac{1}{x}$.

The function is undefined at $x = 0$, so the line $x = 0$ is a vertical asymptote.

For large $|x|$ the term $\dfrac{1}{x}$ approaches zero, so $f(x)$ approaches x. The line $y = x$ is therefore the oblique asymptote.

The graph has two separate branches. Each one lies between the y-axis and the line $y = x$.

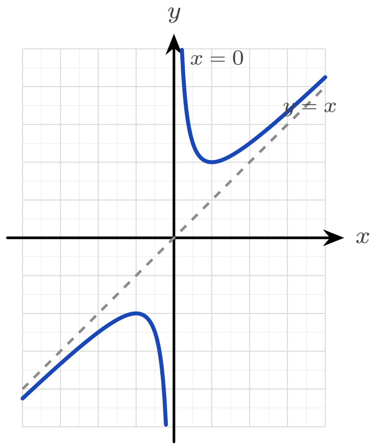

Figure 6.3 The two branches of $y = x + \frac{1}{x}$, each approaching the vertical asymptote $x = 0$ on one side and the oblique asymptote $y = x$ on the other.

In that example the split into a line plus a remainder was already written out for you. In general you must produce it yourself by **polynomial division**.

EXAMPLE 6.3

Find all asymptotes of $f(x) = \dfrac{x^2 - 2x + 3}{x - 1}$.

The denominator is zero at $x = 1$, so there is a vertical asymptote there. The numerator's degree exceeds the denominator's by one, so there is an oblique asymptote, found by dividing:

$$\frac{x^2 - 2x + 3}{x - 1} = x - 1 + \frac{2}{x - 1},$$

since $(x-1)(x-1) + 2 = x^2 - 2x + 3$. As $|x|$ grows the remainder term $\frac{2}{x-1}$ approaches zero, so the curve approaches the line

$$y = x - 1,$$

which is the oblique asymptote.

TIP Show the division, not only the answer. The quotient *is* the asymptote, and the division itself carries most of the marks.

The syllabus also names the reverse shape: a linear numerator over a quadratic denominator, $\dfrac{ax + b}{cx^2 + dx + e}$. Here the denominator has the higher degree, so it grows faster than the numerator and the curve approaches the horizontal asymptote $y = 0$. Each real root of the denominator gives its own vertical asymptote, so this form can have two.

EXAMPLE 6.4

Sketch the key features of $f(x) = \dfrac{x+1}{x^2-4}$.

Factorise the denominator to find the vertical asymptotes: $x^2 - 4 = (x-2)(x+2)$, so there are *two*, at $x = 2$ and $x = -2$.

The numerator has lower degree than the denominator, so the horizontal asymptote is $y = 0$.

For the intercepts, set the numerator to zero to get $x = -1$, and substitute $x = 0$ to get $f(0) = -\frac{1}{4}$.

The sign of f on each interval completes the sketch. The three values $x = -2$, $x = -1$ and $x = 2$ divide the x-axis into four intervals. Test the sign of $\dfrac{x+1}{(x-2)(x+2)}$ on each:

$$x < -2: \ \tfrac{-}{(-)(-)} < 0, \qquad -2 < x < -1: \ \tfrac{-}{(-)(+)} > 0,$$

$$-1 < x < 2: \ \tfrac{+}{(-)(+)} < 0, \qquad x > 2: \ \tfrac{+}{(+)(+)} > 0.$$

Now read the branches from the signs. Where f is positive next to an asymptote the curve rises to $+\infty$; where it is negative it falls to $-\infty$. The middle branch passes through both intercepts.

The rule for reading asymptotes off a rational function is worth stating once for all cases.

Degree of numerator vs denominator	Asymptote as $x \to \pm\infty$
numerator $<$ denominator	horizontal, $y = 0$
numerator $=$ denominator	horizontal, at the ratio of leading coefficients
numerator $=$ denominator $+1$	oblique (a slanted line)

In every case, vertical asymptotes sit at the zeros of the denominator (once common factors are cancelled).

YOUR TURN 6.2 Rational Functions and Oblique Asymptotes HL

6. State the vertical asymptotes of each function.

(a) $f(x) = \dfrac{x^2+1}{x-2}$ [1] (b) $f(x) = \dfrac{1}{(x-1)(x+3)}$ [2]

(c) $f(x) = \dfrac{x}{x^2-9}$ [2] (d) $f(x) = \dfrac{x^2+1}{x^2-4}$ [2]

7. Write each function as a polynomial plus a remainder, and state the oblique asymptote.

(a) $f(x) = \dfrac{x^2+3}{x}$ [2] (b) $f(x) = \dfrac{x^2}{x-1}$ [3]

(c) $f(x) = \dfrac{x^2+1}{x-2}$ [3]

(d) $f(x) = \dfrac{x^2+4x+1}{x+1}$ [3]

8. For each function, state which kind of asymptote it has as $x \to \pm\infty$, and give its equation.

(a) $f(x) = \dfrac{x}{x^2-9}$ [2]

(b) $f(x) = \dfrac{x^2+1}{x^2-4}$ [2]

(c) $f(x) = \dfrac{x^2-9}{x+2}$ [3]

9. Consider $f(x) = x + \dfrac{2}{x}$.

(a) State the vertical asymptote. [1]

(b) State the oblique asymptote. [1]

(c) Explain why the curve never crosses the oblique asymptote. [2]

(d) Sketch the two branches. [2]

6.3 Exponential Functions

An **exponential function** $y = a^x$ places the variable in the exponent. Compare it with a power such as x^3, where the variable is the base and the exponent is fixed. In an exponential the two roles are swapped.

That swap has one important consequence: an exponential changes at a rate proportional to its own size. A quantity that is twice as large grows twice as fast. This is why the same function describes cooling coffee, growing populations and decaying atoms.

For $a > 1$ the graph rises, and rises ever more steeply: this is **growth**. For $0 < a < 1$ it falls towards zero: this is **decay**.

In both cases the graph passes through $(0, 1)$, stays positive everywhere, and approaches the horizontal asymptote $y = 0$ on one side. So the domain of an exponential function is all real numbers, and its range is the positive real numbers, $y > 0$.

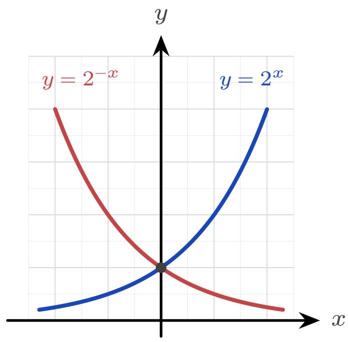

Figure 6.4 Growth and decay are reflections of each other in the y-axis. Both pass through $(0, 1)$ and approach $y = 0$, but on opposite sides.

One base is used more than any other: $e \approx 2.718$. It is the base for which the steepness of the curve equals its height at every point. That property makes the calculus of e^x the simplest of any exponential: its rate of change is again e^x. So every other exponential is usually rewritten in terms of e. See Ch. 14.

KEY · IB Any exponential can be written with base e, and exponential and logarithm undo each other:

$$a^x = e^{x \ln a}, \qquad \log_a(a^x) = x, \qquad a^{\log_a x} = x \quad (a, x > 0,\ a \neq 1).$$

EXAMPLE 6.5

For $f(x) = 5(0.4)^x$, state whether the function grows or decays, and give its domain, range and asymptote.

Look at the base first. Here $a = 0.4$, and $0 < 0.4 < 1$, so the function **decays**.

The factor 5 multiplies every output, so it stretches the graph vertically but changes nothing else. The outputs of $(0.4)^x$ are always positive, so multiplying by 5 keeps them positive.

Hence the domain is all real numbers, the range is $y > 0$, and the horizontal asymptote is $y = 0$.

Transforming an exponential graph

An exponential is rarely met on its own. In a model it usually appears as

$$y = k\,a^x + c,$$

and each constant has a separate effect. The factor k stretches the graph vertically. The constant c moves it up or down.

The vertical shift is the one that changes the asymptote. The graph of a^x approaches $y = 0$, so adding c makes it approach $y = c$ instead. **The horizontal asymptote moves with the graph.**

EXAMPLE 6.6

Sketch $y = 3(2)^x - 4$, stating the asymptote and both intercepts.

Start with the asymptote. The -4 shifts the whole graph down by 4, so the horizontal asymptote is $y = -4$, not $y = 0$.

For the y-intercept, substitute $x = 0$:

$$y = 3(2)^0 - 4 = 3 - 4 = -1.$$

For the x-intercept, set $y = 0$ and solve:

$$3(2)^x = 4 \implies 2^x = \tfrac{4}{3} \implies x = \log_2 \tfrac{4}{3} \approx 0.415.$$

The curve therefore rises from just above $y = -4$ on the left, crosses the axes at $(0, -1)$ and $(0.415, 0)$, and then rises steeply.

CAUTION A shifted exponential does *not* have the asymptote $y = 0$. In $y = k\,a^x + c$ the asymptote is $y = c$. Reading it as $y = 0$ without checking is the most common error in this topic.

Exponential models

Most real quantities are modelled with base e and time as the variable:

$$A(t) = A_0\, e^{kt},$$

where A_0 is the amount at $t = 0$. The sign of k decides the behaviour: $k > 0$ gives growth, and $k < 0$ gives decay. Its size decides the speed.

EXAMPLE 6.7

Japan's population was about 128.1 million in 2010 and about 126.1 million in 2020. Model the population as $P(t) = P_0 e^{kt}$, with t in years after 2010, and use it to estimate the population in 2030.

Take $P_0 = 128.1$, the value at $t = 0$. Now use the 2020 figure to find k. Substitute $t = 10$ and $P = 126.1$:

$$126.1 = 128.1\, e^{10k} \implies e^{10k} = \frac{126.1}{128.1} \implies k = \frac{1}{10} \ln\left(\frac{126.1}{128.1}\right) \approx -0.00157.$$

The value of k is negative, which confirms decay: the population is falling by about 0.16% each year.

For 2030, substitute $t = 20$:

$$P(20) = 128.1\, e^{-0.00157 \times 20} \approx 124.1 \text{ million}.$$

Treat that figure with care. The model assumes the *rate* stays fixed, and Japan's rate of decline is itself increasing, so official projections for 2030 are lower than this. A two-point exponential fit tells you what would happen if nothing changed, which is rarely what happens.

TIP In a modelling question, read A_0 from the value at $t = 0$ before doing anything else. It is usually given directly, and one mark depends on stating it.

YOUR TURN 6.3 Exponential Functions

10. Evaluate each expression.

(a) 2^3 [1] (b) 2^{-2} [1]
(c) e^0 [1] (d) $9^{1/2}$ [1]

11. State the horizontal asymptote and the y-intercept of each graph.

(a) $y = 3^x$ [2] (b) $y = 5 \cdot 2^x$ [2]
(c) $y = 2^x + 5$ [2] (d) $y = 4(3)^x - 1$ [2]

12. State whether each function grows or decays, and give a reason.

(a) $y = (0.5)^x$ [1] (b) $y = 3^x$ [1]
(c) $y = 5e^{-0.2x}$ [2] (d) $y = 2^{-x}$ [2]

13. Solve each equation exactly.

(a) $2^x = 8$ [1] (b) $e^x = 1$ [1]
(c) $5 \cdot 3^x = 45$ [2] (d) $3(2)^x = 4$ [3]

14. Write each in the required form.

(a) 2^x in terms of base e [2] (b) 5^x in terms of base e [2]

15. GDC A population is modelled by $P(t) = 800e^{0.03t}$, with t in years.

(a) State P_0, and say whether the population grows or decays. [2]
(b) Find the population after 10 years. [2]
(c) Find how long it takes the population to reach 1200. [3]

16. GDC A radioactive sample decays according to $A(t) = 50e^{-0.2t}$, with t in days.

(a) Find $A(6)$, correct to three significant figures. [2]
(b) Find the half-life of the sample. [3]

6.4 Logarithmic Functions

Since $y = a^x$ is one-to-one, it has an inverse, and that inverse is the **logarithm** $y = \log_a x$. It answers the reversed question: not "what is a to the x?" but "to what power must I raise a to get x?" Everything about its graph follows from being the mirror image of the exponential in the line $y = x$.

Reflecting swaps the axes' roles. The exponential's horizontal asymptote $y = 0$ becomes the logarithm's *vertical* asymptote $x = 0$; its domain of all reals becomes the logarithm's range, and its range of positive numbers becomes the logarithm's domain. So $\log_a x$ exists only for $x > 0$, passes through $(1, 0)$, and rises without bound, ever more slowly. Its domain is therefore the positive real numbers, $x > 0$, and its range is all real numbers.

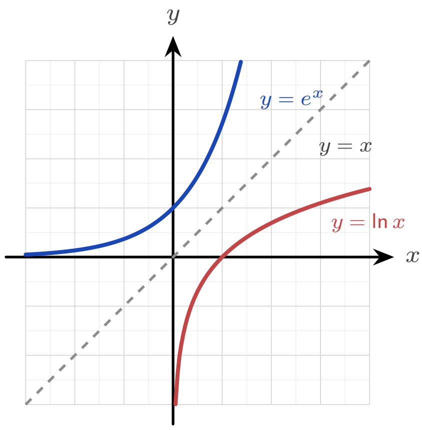

Figure 6.5 The logarithm is the exponential reflected in the line $y = x$. Reflecting exchanges the two axes, which is why one asymptote is horizontal and the other vertical.

CAUTION The logarithm is only defined for positive inputs. Writing $\ln(x-3)$ silently assumes $x > 3$; forgetting this domain restriction is the most common error with logarithmic functions.

Finding the domain

Because a logarithm accepts only positive inputs, every logarithmic function carries a hidden condition. To find the domain, set whatever sits inside the logarithm greater than zero, and solve.

EXAMPLE 6.8

State the domain of $f(x) = \ln(x-3)$ and of $g(x) = \log(5-2x)$.

For f, the input to the logarithm is $x - 3$. Require it to be positive:

$$x - 3 > 0 \implies x > 3.$$

For g, the input is $5-2x$. Require that to be positive:

$$5-2x>0 \implies 5>2x \implies x<\tfrac{5}{2}.$$

Note the direction. Dividing by -2 would have reversed the inequality, so it is safer to move the term across as above.

Transforming a logarithmic graph

The same shifts apply as for any function, but they act on the *vertical* asymptote rather than a horizontal one.

In $y=\ln(x-h)$ the change is inside the bracket, so it moves the graph h units right. The asymptote travels with it, from $x=0$ to $x=h$.

EXAMPLE 6.9

Sketch $y=\ln(x-2)$, stating its asymptote, its domain and its x-intercept.
The input must be positive, so $x-2>0$ and the domain is $x>2$.
That same condition gives the asymptote. As x approaches 2 from above, $x-2$ approaches 0 and $\ln(x-2)$ decreases without bound, so $x=2$ is a vertical asymptote.
For the x-intercept, set $y=0$. A logarithm is zero when its input is 1:

$$x-2=1 \implies x=3.$$

So the curve rises from the asymptote at $x=2$, crosses the axis at $(3,0)$, and continues upward ever more slowly.

Using the inverse relationship

Because the exponential and the logarithm undo each other, you can use a logarithm to solve for a variable in an exponent, and an exponential to solve for a variable inside a logarithm. That is how the inverse of a function built from either one is found.

EXAMPLE 6.10

Find the inverse of $f(x)=3e^{2x}+1$, and state its domain.
Write $y=f(x)$, then undo each operation one at a time until x stands alone:

$$y=3e^{2x}+1 \implies \frac{y-1}{3}=e^{2x}.$$

The variable is now in an exponent, so apply the natural logarithm to both sides:

$$2x=\ln\left(\frac{y-1}{3}\right) \implies x=\tfrac{1}{2}\ln\left(\frac{y-1}{3}\right).$$

Exchange the letters to write the inverse as a function of x:

$$f^{-1}(x) = \tfrac{1}{2} \ln\left(\frac{x-1}{3}\right).$$

Now the domain. The domain of f^{-1} is the range of f. Since $e^{2x} > 0$ for every x, we have $3e^{2x} > 0$ and so $f(x) > 1$. The domain of f^{-1} is therefore $x > 1$, which agrees with requiring $\frac{x-1}{3} > 0$ inside the logarithm.

TIP State the domain whenever a question asks for an inverse involving a logarithm. It is a separate mark, and the quickest check is that the domain of f^{-1} equals the range of f.

YOUR TURN 6.4 Logarithmic Functions

17. Evaluate each expression.

(a) $\log_2 8$ [1]
(b) $\ln 1$ [1]
(c) $\ln e$ [1]
(d) $\log_5 125$ [1]

18. Solve each equation.

(a) $\ln x = 0$ [1]
(b) $\log_3 x = 2$ [1]
(c) $\ln x = 3$ [2]
(d) $\log_2(x-1) = 4$ [2]

19. State the domain of each function.

(a) $y = \ln(x-2)$ [1]
(b) $y = \log(x+5)$ [1]
(c) $y = \ln(2x-6)$ [2]
(d) $y = \log(4-x)$ [2]

20. State the vertical asymptote and the x-intercept of each graph.

(a) $y = \ln x$ [2]
(b) $y = \ln(x+1)$ [2]
(c) $y = \ln(x-2)$ [2]
(d) $y = \log(x+5)$ [3]

21. Find the inverse of each function, and state its domain.

(a) $f(x) = e^x + 2$ [3]
(b) $f(x) = \ln(x-1)$ [3]
(c) $f(x) = 3e^{2x} + 1$ [4]

22. The domain of $y = \ln(9 - x^2)$ is not a single inequality.

(a) Write down the condition that the input must satisfy. [1]

(b) Solve it to find the domain. [3]

6.5 Partial Fractions HL

Adding two simple fractions produces one complicated fraction. **Partial fractions** reverse that process: they split a single fraction back into a sum of simpler ones, one for each factor of the denominator.

The split form is often much simpler to work with than the original. It is also what makes a rational function possible to integrate, so this section is the direct preparation for the integration of rational functions. See Ch. 16.

The method has one precondition, and then three cases decided by the factors in the denominator. Take the precondition first.

The fraction must be proper

A fraction is **proper** when the degree of its numerator is lower than the degree of its denominator. Only a proper fraction can be split into partial fractions.

If the numerator has degree equal to or higher than the denominator, the fraction is **improper**. Divide first, by polynomial division, to write it as a polynomial plus a proper remainder, then split the remainder. For example,

$$\frac{x^2}{x^2-1} = 1 + \frac{1}{x^2-1},$$

and only the proper part $\dfrac{1}{x^2-1}$ is decomposed further.

Case 1: distinct linear factors

This is the only case the AA HL syllabus examines. The denominator is a product of different linear factors, and each factor contributes one term with a constant on top.

KEY For a proper fraction over two distinct linear factors,

$$\frac{px+q}{(x-\alpha)(x-\beta)} = \frac{A}{x-\alpha} + \frac{B}{x-\beta}.$$

To find A and B, multiply through by the denominator, then substitute the values of x that make one factor zero. Each substitution isolates one unknown.

EXAMPLE 6.11

Express $\dfrac{3x+5}{(x-1)(x+2)}$ in partial fractions.

Write

$$\frac{3x+5}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2},$$

and multiply through by $(x-1)(x+2)$:

$$3x+5 = A(x+2) + B(x-1).$$

Setting $x = 1$ isolates A: $8 = 3A$, so $A = \frac{8}{3}$. Setting $x = -2$ isolates B: $-1 = -3B$, so $B = \frac{1}{3}$. Therefore

$$\frac{3x+5}{(x-1)(x+2)} = \frac{8}{3(x-1)} + \frac{1}{3(x+2)}.$$

Choosing the two values that each make one factor zero is what makes the method quick. It turns a pair of simultaneous equations into two one-line calculations.

EXAMPLE 6.12

Express $\dfrac{x+7}{x^2+x-2}$ in partial fractions.

This denominator is not factorised, so factorise it first:

$$x^2 + x - 2 = (x+2)(x-1).$$

Now proceed as before:

$$\frac{x+7}{(x+2)(x-1)} = \frac{A}{x+2} + \frac{B}{x-1} \implies x+7 = A(x-1) + B(x+2).$$

Setting $x = 1$: $8 = 3B$, so $B = \frac{8}{3}$. Setting $x = -2$: $5 = -3A$, so $A = -\frac{5}{3}$. Hence

$$\frac{x+7}{x^2+x-2} = \frac{8}{3(x-1)} - \frac{5}{3(x+2)}.$$

TIP Examination questions usually give the denominator in expanded form. The factorising step is part of the answer and earns a mark, so write it out.

Case 2: a repeated linear factor HL

When a linear factor is squared, one term is not enough. A factor $(x-\alpha)^2$ needs a separate term for each power up to the square: one over $(x-\alpha)$ and one over $(x-\alpha)^2$.

KEY For a proper fraction over a repeated linear factor,

$$\frac{px+q}{(x-\alpha)^2} = \frac{A}{x-\alpha} + \frac{B}{(x-\alpha)^2}.$$

The reason for the second term is short. Placed over the common denominator $(x-\alpha)^2$, the single term $\dfrac{A}{x-\alpha}$ becomes $\dfrac{A(x-\alpha)}{(x-\alpha)^2}$. Its numerator is a multiple of $(x-\alpha)$, so it is zero at $x = \alpha$ and cannot match a general numerator there. The term $\dfrac{B}{(x-\alpha)^2}$ supplies the missing value.

Substituting $x = \alpha$ still isolates the top term, here B. To find A, compare the coefficients of a chosen power of x, since no second root is available.

EXAMPLE 6.13

Express $\dfrac{3x+1}{(x-1)^2}$ in partial fractions.

Write

$$\frac{3x+1}{(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2},$$

and multiply through by $(x-1)^2$:

$$3x+1 = A(x-1) + B.$$

Setting $x = 1$ isolates B: $4 = B$. Comparing the coefficients of x gives $A = 3$. Therefore

$$\frac{3x+1}{(x-1)^2} = \frac{3}{x-1} + \frac{4}{(x-1)^2}.$$

A repeated factor uses substitution for the top term and coefficient comparison for the rest.

Case 3: an irreducible quadratic factor HL

A quadratic factor that does not factorise over the real numbers, meaning its discriminant is negative, is **irreducible**. Over such a factor the numerator is not a constant but a full linear expression $Bx + C$.

KEY For a proper fraction over a linear factor and an irreducible quadratic,

$$\frac{\text{proper}}{(x-\alpha)(x^2+bx+c)} = \frac{A}{x-\alpha} + \frac{Bx+C}{x^2+bx+c}.$$

The numerator over each factor always has degree one less than that factor: a constant over a linear factor, a linear expression over a quadratic. Substituting $x = \alpha$ isolates A; the remaining unknowns come from comparing coefficients.

EXAMPLE 6.14

Express $\dfrac{3x^2-2x+3}{(x-1)(x^2+1)}$ in partial fractions.

The factor x^2+1 has no real root, so it is irreducible. Write

$$\frac{3x^2-2x+3}{(x-1)(x^2+1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+1},$$

and multiply through by $(x-1)(x^2+1)$:

$$3x^2-2x+3 = A(x^2+1) + (Bx+C)(x-1).$$

Setting $x=1$ isolates A: $4=2A$, so $A=2$. Comparing the coefficients of x^2 gives $3=A+B$, so $B=1$. Comparing the constant terms gives $3=A-C$, so $C=-1$. Therefore

$$\frac{3x^2-2x+3}{(x-1)(x^2+1)} = \frac{2}{x-1} + \frac{x-1}{x^2+1}.$$

The three cases follow one rule: give the denominator its factors, then give each factor a numerator of degree one less than itself, adding an extra term for every power of a repeated factor.

Factor in the denominator	Term(s) it contributes
linear $(x-\alpha)$	$\dfrac{A}{x-\alpha}$
repeated linear $(x-\alpha)^2$	$\dfrac{A}{x-\alpha} + \dfrac{B}{(x-\alpha)^2}$
irreducible quadratic (x^2+bx+c)	$\dfrac{Bx+C}{x^2+bx+c}$

Cases 2 and 3 are beyond the AA HL syllabus, which restricts partial fractions to two distinct linear factors.

YOUR TURN 6.5 Partial Fractions HL

23. Express each fraction in partial fractions.

(a) $\dfrac{1}{(x-1)(x-2)}$ [3]

(b) $\dfrac{5}{(x-2)(x+3)}$ [3]

(c) $\dfrac{3}{(x+1)(x+4)}$ [3]

(d) $\dfrac{x}{(x-1)(x+1)}$ [3]

24. Express each fraction in partial fractions.

(a) $\dfrac{2x+1}{x(x+1)}$ [3]

(b) $\dfrac{x+4}{x(x-2)}$ [3]

(c) $\dfrac{2x}{(x-3)(x+1)}$ [3]

(d) $\dfrac{5x-1}{(x-1)(x+2)}$ [4]

25. Factorise the denominator first, then express each fraction in partial fractions.

(a) $\dfrac{4x}{x^2-4}$ [4]

(b) $\dfrac{x+7}{x^2+x-2}$ [4]

26. Consider $\dfrac{x^2}{x^2-1}$.

(a) Explain why this fraction is improper. [1]

(b) Divide to write it as $1+\dfrac{1}{x^2-1}$. [2]

(c) Hence express the whole fraction in partial fractions. [4]

Chapter 6 · Rational and Exponential Functions

Reflections

TOK An asymptote describes a curve approaching a line it never meets, across infinitely many ever-smaller steps. This is the shape of Zeno's ancient paradox: to cross a room you must first cross half, then half of what remains, forever. Motion, Zeno argued, is therefore impossible. Calculus answers him: infinitely many shrinking steps can sum to a finite whole. Does mathematics dissolve the paradox, or merely rename it?

TOK The model in §6.3 fits two population figures and predicts 124.1 million people in Japan in 2030. Official projections are lower, because the rate of decline is itself changing. The model's arithmetic is not wrong. Its assumption is. Every model works by choosing which features of the world to leave out. So when a prediction fails, we have learned something either about the world or about our own choice of what to leave out. How would you tell which?

IM The word *transcendental* was Leibniz's, for quantities that "transcend" algebra, that no polynomial equation can capture. It took until 1873 for Charles Hermite to prove that e is transcendental, and 1882 for Ferdinand von Lindemann to prove the same of π, settling a two-thousand-year-old question: because π is transcendental, the circle cannot be squared with compass and straightedge. Some numbers lie forever beyond the reach of algebra.

RW Exponential decay is the clock of the physical world. Every radioactive isotope falls by half over a fixed *half-life*, from carbon-14's $5{,}730$ years, used to date ancient wood and bone, to isotopes that decay in seconds. The same exponential describes unrestricted population growth and compound interest. When growth is limited by a maximum, it becomes the logistic curve of a spreading rumour or an epidemic. Rational functions model saturation: reaction rates, drug concentrations, and average costs that flatten toward a limit.

EXT The exponential e^x has a *secret identity*. Give it an imaginary exponent and it becomes a rotation: Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$ ties the transcendental e to the trigonometry of Chapter 9 and the complex numbers of Chapter 18. Setting $\theta = \pi$ gives $e^{i\pi} + 1 = 0$, a single line linking e, π, i, 1, and 0, often called the most beautiful equation in mathematics.

End-of-Chapter Problems — Chapter 6: Rational and Exponential Functions

1. Consider $f(x) = \dfrac{2x+3}{x-1}$.

(a) State the vertical and horizontal asymptotes. [2]
(b) Find the x- and y-intercepts. [2]
(c) Sketch the graph. [2]

2. Consider $f(x) = \dfrac{x-2}{x+3}$.

(a) State the domain. [1]
(b) State the asymptotes. [2]
(c) Find $f^{-1}(x)$. [3]

3. Consider $f(x) = \dfrac{x^2-1}{x}$.

(a) State the vertical asymptote. [1]
(b) Find the oblique asymptote. [2]
(c) Find the x-intercepts. [2]

4. Solve $3^{x+1} = 27$. [2]

5. Solve $2^{2x} = 32$. [2]

6. Let $f(x) = e^x$.

(a) Find $f^{-1}(x)$. [2]
(b) State the domain of f^{-1}. [1]

7. Solve $\ln(2x-1) = 0$. [2]

8. Solve $e^{2x} = 5$, giving your answer in exact form. [3]

9. Express $\dfrac{7x-1}{(x-1)(x+2)}$ in partial fractions. [4]

10. Consider $f(x) = \dfrac{3x}{x-2}$.

(a) State the asymptotes. [2]
(b) Show that $f^{-1}(x) = \dfrac{2x}{x-3}$. [4]

11. A bacterial population is modelled by $P(t) = 500 \cdot 2^t$, where t is in hours.

(a) State the initial population. [1]
(b) Find the population after 3 hours. [2]
(c) Find when the population reaches 8000. [3]

12. A radioactive sample has mass $M(t) = 80 \cdot (0.5)^{t/5}$ grams, t in days.

(a) State the initial mass. [1]
(b) State the half-life. [2]
(c) Find the mass after 10 days. [2]

13. Solve $\log_2 x + \log_2(x-2) = 3$. [5]

14. Express $\dfrac{1}{x^2-4}$ in partial fractions. [4]

15. Solve $5 \cdot 3^x = 45$. [2]

16. The value of a car after t years is $V(t) = 20000(0.85)^t$.

(a) State the initial value. [1]
(b) Find the value after 4 years. [2]
(c) State the long-term behaviour of $V(t)$. [1]

17. Consider $f(x) = \dfrac{x^2+x-2}{x-1}$.

(a) Simplify $f(x)$ by factoring. [2]
(b) State the domain. [1]
(c) Describe the graph. [2]

18. Solve $2^x = 3$, giving your answer to 3 significant figures. [3]

19. Given $\log 2 = 0.301$ and $\log 3 = 0.477$, find $\log 6$ and $\log 1.5$. [3]

20. Express $\dfrac{x+5}{x^2+x}$ in partial fractions. [4]

21. Consider $f(x) = \dfrac{4x-1}{2x+1}$.

(a) State the horizontal asymptote. [1]
(b) State the vertical asymptote. [1]
(c) State the y-intercept. [1]

22. Solve $e^{2x} - 3e^x + 2 = 0$. [5]

23. Sketch, on the same axes, $y = 2^x$ and $y = 2^{-x}$, stating the horizontal asymptote and the common point. [4]

24. Solve $\log_3(x+6) = 2$. [2]

25. Show that $f(x) = \dfrac{ax+b}{cx+d}$ is self-inverse (that is, $f(f(x)) = x$) if and only if $a + d = 0$. [6]

26. Solve $4^x - 2^{x+1} = 8$. [5]

27. Express $\dfrac{x^2+1}{x(x-1)(x+1)}$ in partial fractions. [6]

28. Simplify $\log_2 3 \cdot \log_3 8$. [4]

29. The curve $y = \dfrac{x^2+ax+b}{x-1}$ has oblique asymptote $y = x + 3$.

(a) Find a. [3]

(b) Find the value of b for which the curve has a hole at $x = 1$ instead of a vertical asymptote. [3]

30. HL Consider $f(x) = \dfrac{x^2-4x+5}{x-2}$.

(a) Show that $f(x) = (x-2) + \dfrac{1}{x-2}$. [3]

(b) Write down the equations of the vertical and oblique asymptotes. [3]

(c) Find the y-intercept, and show that the curve has no x-intercept. [3]

(d) Show that the curve never meets its oblique asymptote. [2]

(e) Sketch the curve, showing both asymptotes and the y-intercept. [4]

31. GDC The number of people who have heard a rumour t days after it starts is modelled by

$$N(t) = \frac{5000}{1+49e^{-0.8t}}.$$

(a) Find the number of people who had heard the rumour at $t = 0$. [2]

(b) Find $N(5)$, giving your answer to the nearest whole number. [2]

(c) Find the time at which 2500 people have heard the rumour, giving an exact answer and a value to three significant figures. [4]

(d) State the value N approaches as t becomes large, and interpret it in context. [3]

(e) Show that $\dfrac{5000-N}{N} = 49e^{-0.8t}$, and hence explain why plotting $\ln\left(\dfrac{5000-N}{N}\right)$ against t gives a straight line. State its gradient and its vertical intercept. [5]

32. Let $f(x) = \ln(2x - 4)$.

(a) State the domain of f. [2]

(b) Find the exact x-intercept of the graph of f. [3]

(c) State the equation of the vertical asymptote, and describe the behaviour of $f(x)$ near it. [3]

(d) Find $f^{-1}(x)$, and state its domain and range. [4]

(e) Describe a sequence of transformations that maps $y = \ln x$ onto $y = \ln(2x - 4)$. [3]

Challenge Questions

33. **Splitting a sum with partial fractions.** Partial fractions are usually a step towards integration. This investigation uses them for something else: adding infinitely many terms exactly.

(a) Show that $\dfrac{1}{n(n+1)} = \dfrac{1}{n} - \dfrac{1}{n+1}$ for every positive integer n. [2]

(b) Hence write out the sum $\displaystyle\sum_{n=1}^{N} \frac{1}{n(n+1)}$ term by term, and show that almost every term cancels. Deduce that the sum equals $\dfrac{N}{N+1}$. *(Hint: write the first three and the last two terms)* [4]

(c) State the value the sum approaches as $N \to \infty$, and explain what that value means for the graph of the partial sums. [2]

(d) Show that $\dfrac{1}{n(n+2)} = \dfrac{1}{2}\left(\dfrac{1}{n} - \dfrac{1}{n+2}\right)$, and hence find $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n(n+2)}$. [4]

(e) Investigate $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n(n+k)}$ for a positive integer k. Find its value for $k = 3$ and $k = 4$, then state a general formula and justify it. [6]

34. **Newton's law of cooling.** A body at temperature T in a room at temperature R cools so that the difference $T - R$ decays exponentially:

$$T(t) = R + (T_0 - R)\, e^{-kt}, \qquad k > 0,$$

where T_0 is the temperature at $t = 0$ and t is measured in minutes.

(a) Show that the temperature difference $T(t) - R$ falls by the same factor over any interval of fixed length, and that this factor does not depend on T_0. [4]

(b) Deduce that the difference halves every $\dfrac{\ln 2}{k}$ minutes. [3]

(c) A cup of coffee at $90\,^\circ$C is left in a room at $20\,^\circ$C. After 10 minutes it has cooled to $60\,^\circ$C. Find k in exact form, and hence find the half-life of the temperature difference.

[4]

(d) Find, to the nearest minute, how long the coffee takes to reach $40\,^\circ$C. [3]

(e) Explain why the model predicts that the coffee never reaches $20\,^\circ$C, and state what feature of the graph carries that fact. [3]

35. When does an oblique asymptote exist? Consider

$$f(x) = \frac{ax^2 + bx + c}{dx + e}, \qquad a \neq 0,\ d \neq 0.$$

(a) By polynomial division, show that

$$f(x) = \frac{a}{d}\,x + \frac{bd - ae}{d^2} + \frac{r}{dx + e}$$

and find r in terms of a, b, c, d and e. [5]

(b) Hence write down the equation of the oblique asymptote, and explain why the remainder term does not affect it. [3]

(c) Find the condition on a, b, d and e for the oblique asymptote to pass through the origin. [3]

(d) Show that f has no vertical asymptote exactly when $ae^2 - bde + cd^2 = 0$, and explain what the graph does at $x = -\frac{e}{d}$ in that case. [4]

(e) Verify your condition on $f(x) = \dfrac{x^2 + 3x - 4}{x - 1}$, and state what this function really is. [3]